

Space Weather

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Key Points:

- The attentional mechanism is combined with the gated recurrent units (GRU) model to predict the solar wind speed at Lagrangian Point 1 between the Sun and Earth
- Solar wind speed prediction is performed using OMNI and the Solar Dynamics Observatory (SDO) data sets
- Compared with the comparison model, our two-dimensional attention mechanism (TDAM) prediction model achieves the best result 24 h in advance

Correspondence to:

Z. Xie,
caddixie@hotmail.com

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Solar Wind Speed Prediction With Two-Dimensional Attention Mechanism

Yanru Sun¹, Zongxia Xie¹ , Yanhong Chen² , Xin Huang³, and Qinghua Hu¹ 

¹College of Intelligence and Computing, Tianjin University, Tianjin, China, ²National Space Science Center, Chinese Academy of Sciences, Beijing, China, ³National Astronomical Observatories, Chinese Academy of Sciences, Beijing, China

Abstract As more and more high-technical systems are exposed to the space environment, extreme space weather becomes a great threat to human society. In the solar system, space weather is influenced by the solar wind, such that reliable prediction of solar wind conditions in the near-Earth environment effectively reduces the impact of space weather on human society. Solar wind speed prediction is improved by making full use of OMNI data measured at Lagrangian Point 1 (L1) by the National Aeronautics and Space Administration (NASA) and image data observed by the Solar Dynamics Observatory (SDO) satellite in this work. Specifically, we propose a model based on the “**two-dimensional attention mechanism**” (TDAM) to predict solar wind speed. In this study, we first analyze and preprocess data from 2011 to 2017. Second, considering the characteristics of time series data, we adopt the gated recurrent units (GRU) model which can deal with long-term dependence as the prediction part of our model. Third, we design a TDAM, which enables our prediction network to focus on important parts. Three performance indices are used: root-mean-square error (RMSE), mean absolute error (MAE), and correlation coefficient (CC). By comparing TDAM with other models, we find that the TDAM model achieves the best prediction results, with RMSE of 62.8 km/s, MAE of 47.8 km/s, and CC of 0.789 24 h in advance. The experimental results show that the proposed TDAM model can improve the prediction accuracy of solar wind speed.

Plain Language Summary The solar wind refers to the flow of charged particles emitted from the upper atmosphere of the Sun, which fills the whole solar system and has an important impact on Earth's atmosphere. This work focuses on forecasting the parameters of the solar wind using a machine learning model. Solar images in various radiation bands collected with the Solar Dynamics Observatory (SDO) are input as information sources, as well as decades of solar wind data from a region just sunward of Earth. A deep learning model based on the two-dimensional attention mechanism (TDAM) is designed to predict the solar wind. Data gathered in 5 years are used to train the parameters of the model. Good performance is observed when comparing other current techniques.

1. Introduction

“The term space weather refers to conditions on the Sun and in the solar wind, magnetosphere, ionosphere, and thermosphere that can influence the performance and reliability of space-borne and ground-based technological systems and that can affect human life and health (the definition used by the U.S. National Space Weather Plan)” (Schwenn, 2006). As more and more high-tech systems are exposed to the space environment, space weather has an increasing impact on human beings (Hapgood, 2011). For example, the navigation performance and positioning accuracy of the Global Position System (GPS) and BeiDou Navigation Satellite System is affected by the electromagnetic environment in space. Severe space weather activities may lead to failure of these systems (Bedingfield, 1996). In the solar system, space weather is mainly influenced by solar wind and interplanetary coronal mass ejection (ICME), the interplanetary counterparts of coronal mass ejection (CME). Therefore, the establishment of a reliable solar wind prediction model can reduce the harm caused by space weather to human society, and it is also of great significance for the establishment of an effective space weather prediction system.

Over the years, various methods have been proposed to predict features of the solar wind. Generally speaking, three main methods are used: manual monitoring, the physics-based models, and artificial intelligence

based models. At first, some experts observe the Sun's activity in real-time and use their domain knowledge to judge the state of the near-Earth solar wind (Zwickl et al., 1998), combined with the solar wind speed measured by the WIND and ACE satellites located at Lagrangian Point 1 (L1) (King & Papitashvili, 2005; Tylka, 2001). Then, the physics-based models are developed. Yang et al. (2018) review solar wind speed prediction using the advance magnetohydrodynamic (MHD) models, empirical and semiempirical models. A large number of MHD models have been developed by researchers such as the ENLIL model (Odstrcil, 2003) used in the Space Weather Prediction Center of the National Oceanic and Atmospheric Administration (SWPC/NOAA). Empirical models and semiempirical models are also important methods for solar wind speed prediction. The Wang-Sheeley-Arge/Heliospheric Upwind eXtrapolation (WSA/HUX) model (Arge et al., 2003; Reiss et al., 2020) evolves the solar wind conditions from near the Sun to the Earth. For WSA model, a root-mean-square error (RMSE) of 99.5 km/s and a Pearson's correlation coefficient (CC) of 0.35 are reported by Reiss et al. (2016) for 2011–2014. Other important empirical models detect the size of the solar corona area on the surface of the Sun, the current state of the Sun is judged according to the non-linear relationship between these features and the solar wind speed (; Odstrcil, 2003; Odstrcil et al., 2004; Wang & Sheeley, 1990). In recent years, artificial intelligence based models have been designed as we have enough data for training effective models. These models extract relevant features and then use support vector machines (SVM) or artificial neural networks to make predictions. The Potential Field Source Surface (PFSS) model (Schatten et al., 1969) is usually used to calculate the coronal magnetic field and derive the expansion factor (f_s) of it at the source surface (2.5 solar radii [RS]) and the angular distance from the coronal hole boundary (Θ_b). Yang et al. (2018) propose a hybrid intelligent source surface model (HISS), using the output of the PFSS model as the input of artificial neural network to predict the solar wind speed four days in advance, and obtained a correlation coefficient of 0.74. Recently, Upendran et al. (2020) use the extreme ultraviolet (EUV) image combine with deep learning method to predict the solar wind speed, and realizes that the prediction correlation coefficient of 0.54 and RMSE of 80.27 km/s for the daily solar wind speed data 1 day in advance.

However, the existing artificial intelligence methods (Upendran et al., 2020; Yang et al., 2018) assume that different features are equally important for solar wind speed prediction. What's more, these methods cannot introduce long-term information of features for prediction (Bussy-Virat & Ridley, 2014; Liu et al., 2011). In fact, there's an implicit relationship between the solar wind speed and the data of different features at different moments (Krieger et al., 1973; Rotter et al., 2012, 2015). These features such as coronal holes and CME can sustain for a period of time and influence the solar wind speed after reaching L1 point.

In order to solve the upper two problems, attention mechanism and gated recurrent units (GRU) in machine learning are introduced in our model. Attention mechanism (Yohanandan et al., 2018) is a brain signal processing mechanism in human vision. By quickly scanning the global text or image, the human can focus on areas that need more attention, greatly improving the efficiency of selecting high-value information from the vast amount of data. The attention mechanism is introduced in deep learning (Bahdanau et al., 2014) and the core goal is to select the information which is critical to the current task by adjusting the parameters in the network. Therefore, it can help the neural network to capture key features that have greater influence on the prediction results and improve the prediction effectiveness.

GRU is proposed by Cho et al. (2014). It is a kind of recurrent neural network (RNN) (Rumelhart et al., 1986) model. RNN is one of the typical models used in deep learning (LeCun et al., 2015; Schmidhuber, 2015). Long short-term memory (LSTM) (Elman, 1990) improves the RNN model. The LSTM unit consists of three kinds of gates: the input gate, the forgetting gate, and the output gate, which establish a self-cycling of the internal state of the LSTM unit. LSTM reduces gradient disappearance and gradient explosion by introducing gate operation, making it able to deal with both short-term dependence and long-term dependence problems. The GRU model has become a widely used timing prediction model by combining forgetting gate which determines, what information should be discarded or retained and input gate in LSTM into a single "update gate," which reduces the number of parameters, avoids overfitting.

In this work, we combine GRU with the attention mechanism, and propose a prediction model based on the "two-dimensional attention mechanism" (TDAM) to predict the solar wind speed at the L1 point. We first calculate the sum of the pixel values of coronal holes on the surface of the Sun by threshold segmentation to represent the coronal hole information. And then, it is used as input together with solar wind historical

features, ICME list, and other time-series data into the TDAM model. Through the TDAM, our network has the ability to capture long-term dependent features and pay more attention to features and moments which have a larger impact on the results. GRU is used to excavate the implicit relation between solar wind speed and coronal hole, ICME and other influencing features. The results of solar wind speed prediction 24 h in advance are RMSE: 62.8 km/s, MAE: 47.8 km/s, CC: 0.789.

The rest of this study is structured as follows: Section 2 carries out data analysis and processing, including data selection, data processing, and evaluation indicators. In Section 3, we introduce the structure of the prediction model based on TDAM, including GRU and attention mechanism. Section 4 provides the prediction results of the TDAM model and comparative experiments. Section 5 summarizes the work.

2. Data Analysis and Processing

2.1. Data Selection

We collect the OMNI data set released by NASA (<https://omniweb.gsfc.nasa.gov/index.html>) and the ICME list in [http://www.srl.caltech.edu/ACE/ASC/DATA/level3/icmetable2.htm#\(a\)](http://www.srl.caltech.edu/ACE/ASC/DATA/level3/icmetable2.htm#(a)). Besides, we collect the data set of EUV filtergrams from the SDO (Pesnell et al., 2011) satellite, courtesy of NASA/SDO and the Atmospheric Imaging Assembly (AIA), EVE, and HMI science teams.

The OMNI data set is a collection of in situ measurements of plasma and magnetic field properties by a number of spacecraft orbiting the L1 point between the Sun and Earth. Since we predict the solar wind speed 24 h in advance, the OMNI data set with a resolution of 1 h is selected in this study. The target output of the prediction model is the hourly mean solar wind speed measured at L1 after 24 h.

The SDO satellite is a NASA mission which has been observing the Sun since 2010 (Bourkland & Liu, 2011). Launched on February 11, 2010, the observatory is part of the Living With a Star (LWS) program. The AIA (Lemen et al., 2011), led by the Lockheed Martin Solar and Astrophysics Laboratory (LMSAL), consists of four telescopes that employ normal-incidence and provides continuous full-disk observations of the solar chromosphere and corona in seven EUV channels. It also provides unprecedented views of the various phenomena that occur within the evolving solar outer atmosphere. Since the solar wind speed is also affected by the phenomenon of CMEs, the ICME list is chosen to represent its features. The near-Earth ICMEs, which are the interplanetary counterparts of CMEs occurring near the Sun, have been well documented in a list by Richardson and Cane (2015).

To test the generalization of the TDAM model for solar wind speed prediction and due to the availability of data, the data from 2011 to 2017 are sampled every hour. The data from 2011 to 2015 are used as a training set, consisting of 43,824 samples. 8,784 samples in 2016 are used as a validation set. Finally, 8,760 samples in 2017 are used as a test set.

2.2. Data Processing

2.2.1. Input Features

Temmer et al. (2018) discover the relation between coronal hole and solar wind speed. Simultaneously, similar relations are observed between solar wind features and geomagnetic effects (Rotter et al., 2015). We select five solar wind features from the OMNI data set as our input features, namely, Bulk Speed, Proton density, Proton temperature, Flow Pressure, and Sigma-B. Sigma-B is the length of the vector formed by the standard deviation of the mean of the components:

$$\text{Sigma} - B = \sqrt{\sigma(B_x)^2 + \sigma(B_y)^2 + \sigma(B_z)^2}, \quad (1)$$

Since the solar wind is mainly affected by coronal holes and ICME, in addition to considering the historical solar wind features, we introduce the features of these two phenomena to assist the training of our model. We use images of AIA 131 Å and extract information of coronal holes by using a threshold-based method. The ablation experiment (He et al., 2017) in Section 4.2 prove that the best results could be obtained by setting the threshold to 15. Specifically, after converting AIA 131 Å images of the solar surface into

Table 1

A List of all the Time Series Data we Used

	Parameter	Symbol	Cadence	Time
OMNI data set	Proton density	Density	1 h	2011.1–2017.12
	Proton temperature	Temperature	1 h	2011.1–2017.12
	Flow pressure	Pressure	1 h	2011.1–2017.12
	Sigma-B	Sigma-B	1 h	2011.1–2017.12
	Bulk speed	Speed	1 h	2011.1–2017.12
ICME list	Starting state	Start_state	1 h	2011.1–2017.12
	End state	End_state	1 h	2011.1–2017.12
AIA 131	Coronal hole information	Coronal hole	1 h	2011.1–2017.12

Abbreviations: AIA, atmospheric imaging assembly; ICME, interplanetary coronal mass ejection.

grayscale images, the sum of pixel values less than 15 in the images is calculated to represent coronal hole information:

$$\text{coronal hole information} = \sum_{i=0}^m \sum_{j=0}^n A_{i,j}, \quad (2)$$

$$A_{i,j} = \begin{cases} A_{i,j}, & A_{i,j} \leq 15 \\ 0, & A_{i,j} > 15 \end{cases} \quad (3)$$

where m and n are the sizes of the images, and $A_{i,j}$ is the pixel value of the image of the Sun at each moment. Since the average time for the solar wind traveling from the Sun to L1 is 96 h (Yang et al., 2018), the features observed in the solar image taken at current will affect the solar wind speed at L1 after 96 h. As the speed of the solar wind is different, the travel time is also different. So the coronal hole features between $t - 106$ and $t - 86$ before the prediction point t are selected as the input.

As ICMEs are plasma ejected from the Sun's corona and have a great influence on the solar wind speed, we introduce the ICME features according to the ICME list information. To be specific, binarization of ICME list is converted into a new feature, which is 1 for the corresponding moment of ICMEs and 0 for other moments.

$$ICME_t = \begin{cases} 1, & \text{if a ICME occurs at time } t \\ 0, & \text{if there is no ICME at time } t \end{cases} \quad (4)$$

To sum up, we select seven related features as the input of the model and prove all these features achieve the best performance in Section 4.1. The solar wind speed measured at L1 point after 24 h is used as the output of the model. The input and output features are shown in Table 1.

2.2.2. Outlier Processing and Data Normalization

As the observation equipment may sometimes fail or be in abnormal states, there will be outliers and missing instances in both time series data and image data. Since the gap of missing data is less than 12 h, so if a feature value is missing, we estimate it with the neighboring values by the cubic B-spline curve interpolation (Boehm, 1980).

$$P(t) = \sum_{i=0}^n P_i F_{i,k}(t), \quad (5)$$

where P_i is the feature point of the control curve, and $F_{i,k}(u)$ is the B-spline basis function of order k , and

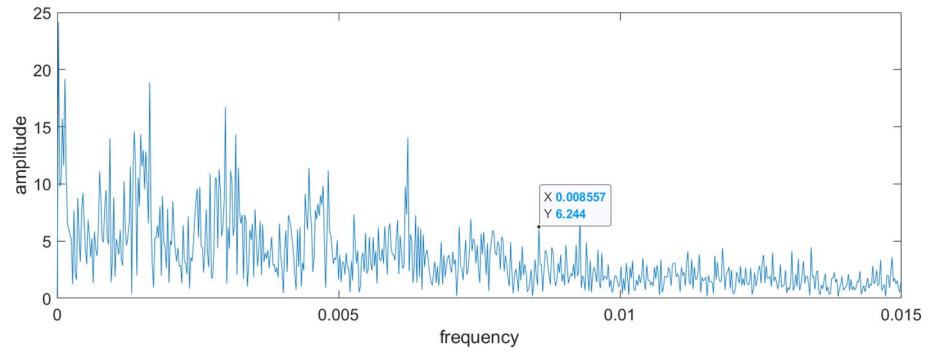


Figure 1. Solar wind speed frequency-amplitude curve from 2011 to 2015.

$$F_{i,k}(t) = \frac{1}{k!} \sum_{l=0}^{k-i} (-1)^l \binom{l}{k+1} (t+k-l-f)^k, \quad (6)$$

where $\binom{l}{k+1}$ stands for factorial.

The solar image may be missing or unclear. Coronal holes which are the primary source of the fast solar wind are fairly stable, thus the images taken in the neighboring time window are almost the same. So images near the time in the SDO high frequency data set are used to replace them. For example, if the image at 10:00 on January 1, 2011 is missing, we will use the image at 10:01 or 9:59 in the high-frequency data. If the images of these two moments are still missing, we will look for the images near these two moments again in the same way, so as to generate clean data and facilitate the extraction of subsequent features.

Different evaluation indexes often have different dimensions and dimensional units, which affects the results of data analysis. In the field of machine learning, data standardization is used to eliminate the dimensional influence between different features. After standardized processing of the original data, each feature is in the same order of magnitude, which is suitable for comprehensive comparative evaluation. The standard deviation of the data in this study is standardized according to the mean and standard deviation of the original data. The processed data conforms to the standard normal distribution, that is, the mean value is 0 and the variance is 1. For each feature, the formula is: $x^* = \frac{x - \mu}{\sigma}$, where x represents the sample data, x^* represents the normalized data, μ is the mean of all sample data, and σ is the standard deviation of all sample data. In machine learning, each feature is normalized according to its mean and variance, respectively for eliminating the dimensional influence.

2.2.3. Periodic Calculation and Data Alignment

We treat each time point in the time series as a dimension, so our data lie in a high-dimensional space. In the time domain, aggregation in different time units is the simplest time series feature, but this method does not consider the displacement information of the time series signal. Another method to extract the features of time series is through frequency domain transformation. The features in frequency domain may reduce the dimension of feature space and make the feature more compact. Since the data we use are discrete sampled, we use the discrete Fourier transform (Weinstein & Ebert, 1971) to convert the time domain signal to the frequency domain signal. The sequence $\{x_n\} := x_0, x_1, \dots, x_N$ is transformed into the new sequence $\{X_d\} := X_0, X_1, \dots, X_N$ by the formula

$$X_d = \sum_{n=0}^N x_n \cdot e^{-\frac{2\pi i}{N} kn}, \quad (7)$$

where $d = 0, 1, 2, \dots, N - 1$ and the frequency information in the time domain signal is extracted. The corresponding spectrum diagram is shown in Figure 1. If the frequency selection is too low, the period is too

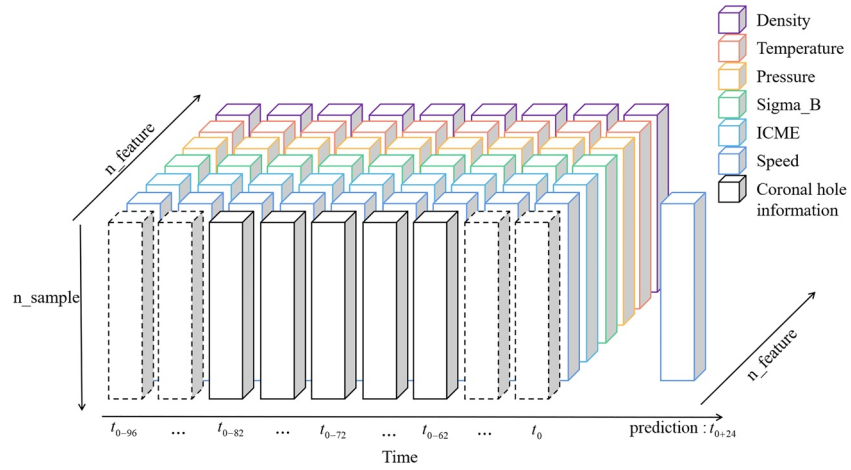


Figure 2. Data alignment of the two-dimensional attention mechanism (TDAM) prediction model. Different colored bars represent different features. The black dotted line column represents the coronal hole features, and we add zeros to its time-domain signal to increase its length for data alignment. We use data from 96 h ahead of the current moment to predict the wind speed 24 h after the current moment.

long, the memory required is too large, and the training model takes a long time. If the frequency selection is too high and the period is too short, the model cannot obtain enough information to make an accurate prediction. Therefore, after many experiments, we choose a frequency of 0.0085 to achieve a good balance between time consumption and accuracy. The period is 120 h.

As to the issue of time series prediction, there exists the situation of data alignment. The input features listed in Table 1, except the coronal hole information, adopted the data L hours before the prediction point. For EUV images, we remove the interference of coronal hole information in other less influential periods, and further accurately, the features we select. For the other data, we consider features long enough as input. After experimental verification, we found that $L = 96$ achieved the best experimental performance. The data alignment of the “TDAM” model proposed by us is shown in Figure 2.

2.3. Assessment Criteria

In the process of model prediction, we should introduce some metrics to quantify the performance. We calculate the RMSE, mean absolute error (MAE), and CC between the predicted value and the observed value, defined as (Yang et al., 2018):

1. Mean absolute error:

$$MAE = \frac{\sum_{k=1}^N |o^k - p^k|}{N}, \quad (8)$$

2. Root mean square error:

$$RMSE = \sqrt{\frac{\sum_{k=1}^N (o^k - p^k)^2}{N}}, \quad (9)$$

3. Correlation coefficient:

$$CC = \frac{\sum_{k=1}^N (p^k - \bar{p})(o^k - \bar{o})}{\sqrt{\sum_{k=1}^N (p^k - \bar{p})^2} \sqrt{\sum_{k=1}^N (o^k - \bar{o})^2}}, \quad (10)$$

where o^k is the k th observed speed value, p^k is the k th predicted speed value, $\bar{o}$ and $\bar{p}$ are the mean value, and N is the batch_size of data.

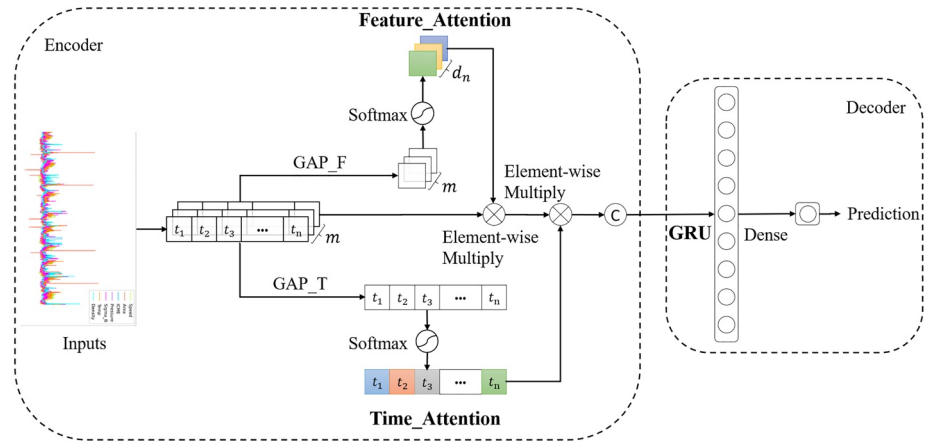


Figure 3. Two-dimensional attention mechanism (TDAM) model. The input is preprocessed time series data, d_n refers to the feature dimension, and t_n refers to the time dimension, GAP refers to global average pooling.

These metrics evaluate the model from different perspectives. MAE is the average of absolute error, which can better reflect the actual situation of the predicted error. RMSE uses average error, is sensitive to very large or very small errors in a set of measurements, and is a good indicator of the accuracy of the predicted value. RMSE and MAE are used to measure the deviation between the observed and the predicted values, and eliminate the influence of sample size N and dimension. But, they cannot accurately express the correlation degree between the predicted value and the observed value, so Pearson correlation coefficient CC is used to measure the linear correlation between the predicted value and the observed value. The smaller RMSE and MAE, the larger CC, and the better the prediction is.

3. The Proposed Model of TDAM

In this section, we introduce the proposed model for predicting the solar wind speed. The TDAM model uses the encoder-decoder architecture (Cho et al., 2014; Bahdanau et al., 2014). The encoder extracts the attention from the characteristic dimension and the time dimension of the input data by using the two-dimensional attention mechanism, and multiplies them with the input data. Then the decoder uses GRU to predict the solar wind speed 24 h in advance. The overall model structure is shown in Figure 3.

There are two main parts in the TDAM model: encoder and decoder. For the encoder part, we introduce the attention mechanism. The task of attention mechanism is to highlight the focus. We propose the TDAM to give greater weight to more important features and moments. First, we extract the important feature which is helpful to improve prediction through the global average pooling (GAP) (Lin et al., 2013) and softmax function.

$$Feature_attention = softmax(GAP_F(X)) = softmax\left(\frac{\sum_{i=0}^t x_{i,m}}{t}\right), \quad (11)$$

Feature_attention gives more parameters to the more important features, such as speed. Then, we use a similar method to extract the important moment.

$$Time_attention = softmax(GAP_T(X)) = softmax\left(\frac{\sum_{j=0}^m x_{t,j}}{m}\right), \quad (12)$$

Through Time_attention, the more important moment obtain more parameters such as the moments near the current time. Finally, these attention are multiplied by the input data X to enhance them and obtain the intermediate semantic C :

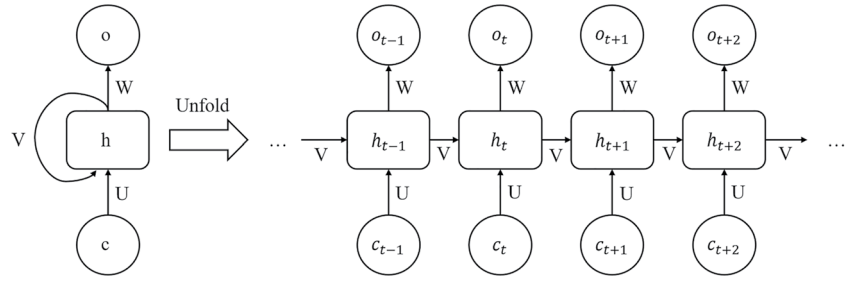


Figure 4. Unfolding network of recurrent neural networks through time.

$$C = X * Feature_attention * Time_attention, \quad (13)$$

The TDAM can enable the network to automatically learn the important part of the time dimension and feature dimension and allocate more attention for them to achieve better prediction. In Section 4.2, we also carry out ablation experiments to prove the effectiveness of our TDAM.

For the decoder part, we use GRU to decode the intermediate semantic C generated by the encoder and predict the speed. Because RNN increases the time dimension, they are effective in solving timing prediction tasks. Here, we compare three different models: basic RNN, LSTM, and GRU. Figure 4 shows the expanded structure of basic RNN over time. For the basic RNN, due to the gradient disappearance problem, it can deal with some short-term dependence, but cannot deal with long-term dependence problems. LSTM reduces the possibility of gradient disappearance and gradient explosion by introducing gate operation, making it deal with both short-term dependence problems and long-term dependence problems. GRU combines the forget gate f_t and input gate i_t in LSTM into a single update gate z_t , which has fewer parameters, faster training speed and less overfitting. Therefore, we choose GRU as the decoder part of the TDAM model for training, verification, and testing. Figure 5 is the internal structure of GRU. Equations 14–17 are the calculation method of GRU (Cho et al., 2014):

$$z_t = \sigma(W_z \cdot [h_{t-1}, c_t]), \quad (14)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, c_t]), \quad (15)$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, c_t]), \quad (16)$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t, \quad (17)$$

where $(\cdot)$ represents the connection of two matrices, $*$ represents the product of matrices, σ is the sigmoid function, h_{t-1} is the state transmitted at the last moment, and c_t is the input of the current node. z_t is the update gate, deciding what information to discard and what new information to add. r_t is the reset door,

used to determine how much past information to forget. $\tilde{h}_t$ is the candidate hiding layer, representing the new information at the current moment. In Section 4.2, we also carry out ablation experiments to compare the performance of three decoders.

The training algorithm of the TDAM model is backward propagation over time. The input x_i of TDAM at each moment is a time series data with length L , $x_i \in R^{T \times m}$, where m is the number of attributes. Therefore, the input to the model is a three dimensional matrix, $x \in R^{N \times T \times m}$, where N is the size of a batch_size data.

We also conduct a lot of experiments to determine the appropriate input length. Finally, we select the input length $L = 96$ in combination with the period calculated in Section 2.2. As one can see from the comparison

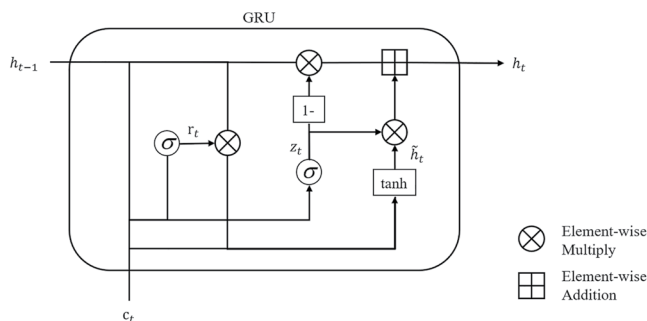


Figure 5. Internal structure of gated recurrent units (GRU).

Table 2
Performance Comparison Between Two-Dimensional Attention Mechanism (TDAM) and Benchmark Models

Model	RMSE	MAE	CC
Persistence	76.050	57.720	0.719
AM	106.882	82.825	0.001
SVR_Linear	75.279	52.985	0.719
SVR_Poly	76.465	53.255	0.707
SVR_RBF	75.396	53.098	0.720
LSTM	81.164	60.798	0.614
GRU	87.804	64.027	0.539
TDAM_S	68.471	52.935	0.742

Notes. Bold values are the best performance compared with the other methods in each table.

Abbreviations: AM, autoregressive model; CC, correlation coefficient; GRU, gated recurrent units; LSTM, long short-term memory; MAE, mean absolute error; RMSE, root-mean-square error; SVR, support vector regression; TDAM, two-dimensional attention mechanism.

experiment in Section 4.2.2, this input length setting improves the performance compared to the other settings. Now, we confirm the input parameters of the model, the TDAM model takes a time series of speed, proton density, proton temperature, flow pressure, sigma-B, ICME, and coronal hole information between $t - 120$ and $t - 24$ as input. The output of the model is the solar wind speed at the moment ($t + 24$), where t is the current time.

The experiments in this study use the RTX 2080 server, python3.7. Batch_size is 128, the optimizer is Adam, the learning rate is set to 0.1, and the gradient clipping value is set to 5.0. The learning rate is monitored through ReduceLROnPlateau. If the model effect is not improved after five epochs iterations, that is, the loss of the verification set is not reduced, the learning rate is multiplied by 0.1 to continue the training. A total of 50 epochs are executed.

4. Experiments and Results

In this section, we present the results of contrast experiments for solar wind speed prediction: persistence model (Owens et al., 2013; Reiss et al., 2016), autoregressive model (Box & Jenkins, 1970; Poggi et al., 2003), support vector regression (Drucker et al., 1996), only-LSTM,

and only-GRU, and ablation experiments. In Section 4.1, we introduce the comparison models and present the results of them and the TDAM model. In Section 4.2, a number of ablation experiments are conducted, including the combination of different input features, length selection of input feature, threshold selection of coronal hole features, setting of attention mechanism, and decoder.

4.1. Benchmark Models

We describe the contrast models in this subsection. The comparison models adopt speed as input and carry out the same processing as the TDAM model (TDAM_S), so the comparison is reasonable.

The first baseline method is the **Persistence Model** (Owens et al., 2013), which is one of the easiest ways to predict time series. We input the solar wind speed at time $t - 24$ into Persistence Model and get the speed at current time: $V_t = V_{t-24}$, where $t_H = 24$.

Autoregressive Model (AM) (Box & Jenkins, 1970; Poggi et al., 2003) is a statistical method for processing time series, using the data from $t - h - 1$ to $t - 1$ as input to predict the current solar wind speed V_t ($V_t = c + \sum_{i=t-h-1}^{t-1} \alpha V_{t-i}$). The model assumes that there is a linear relationship between the input and output, and the predicted value at time t is calculated by regression equation. This method is simple and efficient, and is a common method for time series prediction.

SVM is widely used in pattern recognition and function estimation. **Support vector regression (SVR) algorithm** (Drucker et al., 1996) is the application of SVM in function approximation and regression estimation. In SVR regression analysis, the use of SVM can make the regression function as smooth as possible, with strong generalization ability. We use three kernel functions in this model: linear kernel function, polynomial kernel function, and radial basis kernel function. In this work, we adopt SVR implemented by Scikit-learn (Pedregosa et al., 2011) to conduct experiments.

In addition, we compare TDAM with **LSTM** and **GRU**. As shown in Table 2, we summarize the experimental results of our model and other benchmark models based on the performance indices: RMSE, MAE, and CC. We can see that our model achieves the best performance, with the lowest RMSE and MAE, and the highest CC. It can also be seen from Figure 6 that our model achieves the best fitting of the observed values and better captures the upward and downward trend of the solar wind speed. The reason is that the TDAM learn the feature dimension and the time dimension automatically.

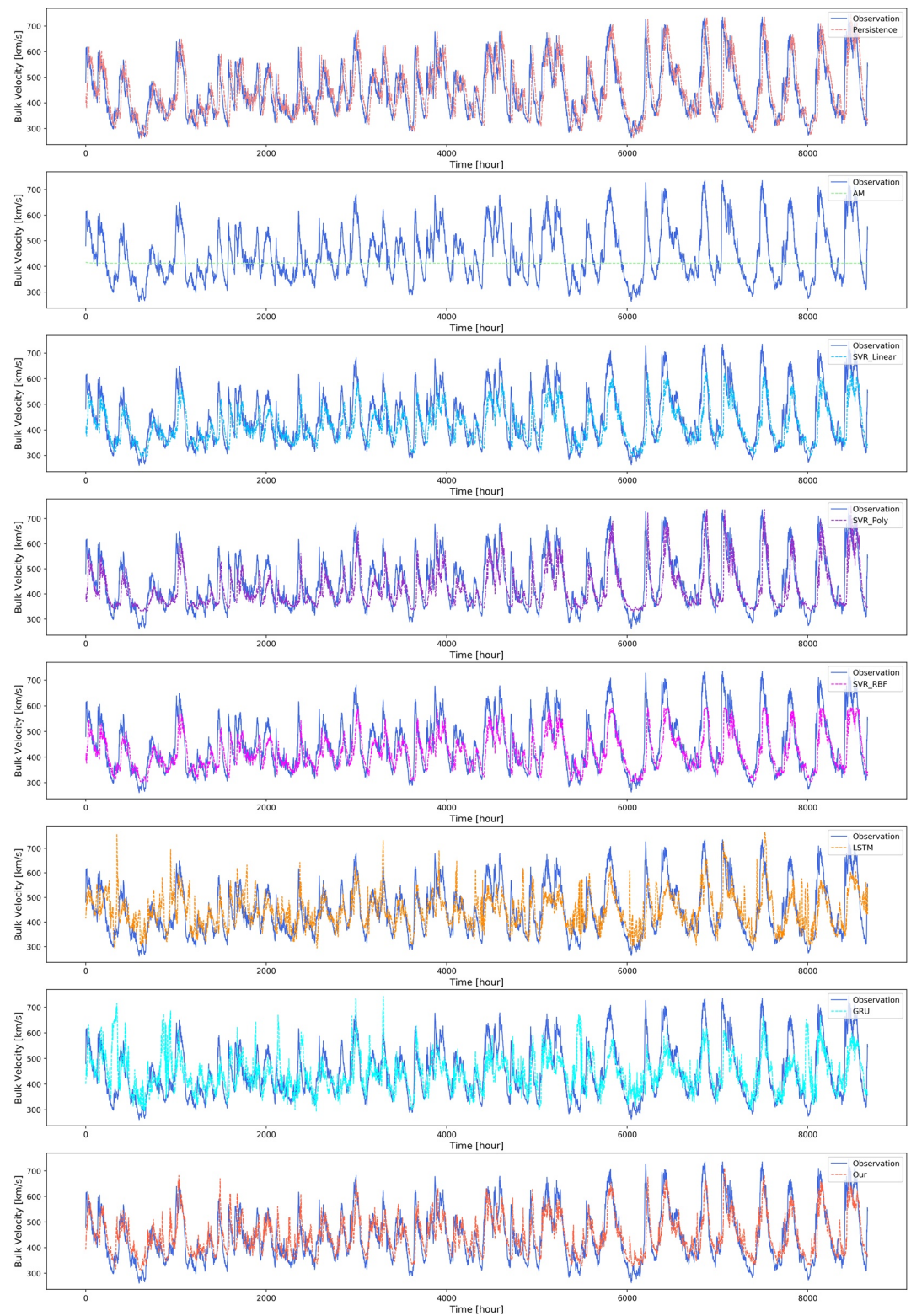


Figure 6. Fitting curves of the predicted and observed values of all the benchmark models and the TDAM model. From top to bottom are the persistent model, autoregressive model, support vector regression (SVR)_Linear model, SVR_Poly model, SVR_RBF model, long short-term memory (LSTM), gated recurrent units (GRU) and our two-dimensional attention mechanism (TDAM) prediction model, respectively.

Table 3
Performance Comparison Between Different Combinations of Input Parameters and Two-Dimensional Attention Mechanism (TDAM)

Input_Feature	RMSE	MAE	CC
Speed	68.471	52.935	0.742
Speed + Coronal hole	68.298	53.330	0.745
Speed + Coronal hole + Density	65.276	49.922	0.773
Speed + Coronal hole + Density + Tempure	65.377	49.518	0.771
Speed + Coronal hole + Density + Tempure + sigma_B	65.242	49.758	0.772
Speed + Coronal hole + Density + Tempure + sigma_B + Pressure	65.538	50.201	0.767
TDAM	62.882	47.814	0.789

Notes. Bold values are the best performance compared with the other methods in each table.

Abbreviations: CC, correlation coefficient; RMSE, root-mean-square error; MAE, mean absolute error; TDAM, two-dimensional attention mechanism.

4.2. Ablation Experiments

4.2.1. Selection of Features and Input Length

We try the combination of different input features, and the results are shown in Table 3. It can be seen that our prediction accuracy improves with the increase of the integrated features. It shows that there is an implicit relation between different input features and solar wind speed, and our model can learn this implicit relation well. In the end, we get the best results for the experiment that incorporates all the features.

Figure 7 gives the model parameters, and the parameter proportions of each feature varies at different time. Our model can well learn the implicit relationship between them. It can be seen that the coronal hole information gets more attention allocation around 24 h, which is the same as previous analysis.

4.2.2. Selection of Input Length

In addition to the input features, the input length of data also has a very important effect on improving the prediction accuracy. Through several groups of comparative experiments in Table 4, it can be found that if the input length is too short, then the input information will be too less to make accurate predictions. However, if the input data are too long, there will be more redundant information, which will also affect the accuracy of the prediction. Combined with the period calculation in Section 2.2, $L = 96$ is selected as the length of the input data and the best prediction result is achieved.

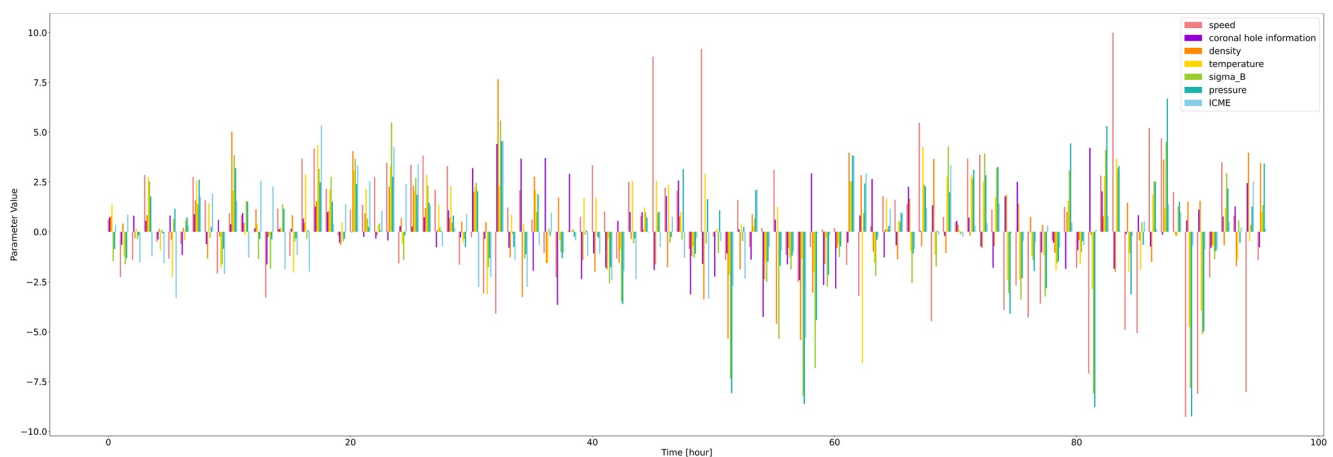


Figure 7. The parameter value of the model (attention allocation) when the input data dimension is [96, 7]. Each moment contains the parameter values of seven features, representing the attention allocation of the model to different features at this moment. The x-axis is 96 moments, the y-axis is the obtained parameter, and different features are shown in different colors.

Table 4
Performance Comparison Between Different Input Length Settings and Two-Dimensional Attention Mechanism (TDAM)

Input_Length	RMSE	MAE	CC
84 h	63.337	48.396	0.785
96 h (our)	62.882	47.814	0.789
108 h	63.973	48.662	0.780
120 h	64.144	49.269	0.779
132 h	63.373	48.651	0.781
144 h	65.920	49.969	0.763

Notes. Bold values are the best performance compared with the other methods in each table.

Abbreviations: CC, correlation coefficient; RMSE, root-mean-square error; MAE, mean absolute error; TDAM, two-dimensional attention mechanism.

Table 5
Performance Comparison of Different Coronal Hole Information Threshold Settings and the Two-Dimensional Attention Mechanism (TDAM) Model Setting

Coronal hole information threshold	RMSE	MAE	CC
10	65.967	50.047	0.764
15 (our)	62.882	47.814	0.789
20	64.932	49.279	0.773

Notes. Bold values are the best performance compared with the other methods in each table.

Abbreviations: CC, correlation coefficient; MAE, mean absolute error; RMSE, root-mean-square error.

Table 6
Performance Comparison Between Different Attention Mechanism Settings and Two-Dimensional Attention Mechanism (TDAM)

Attention mechanism	RMSE	MAE	CC
no_attention	67.100	51.137	0.754
time_attention	64.208	48.477	0.782
feature_attention	73.517	54.990	0.699
feature_time_attention (our)	62.882	47.814	0.789

Notes. Bold values are the best performance compared with the other methods in each table.

Abbreviations: CC, correlation coefficient; MAE, mean absolute error; RMSE, root-mean-square error.

4.2.3. Selection of Coronal Hole Information Threshold

The size of the coronal hole has an important effect on the speed of the solar wind: the larger the coronal hole, the more particles ejected, the stronger the solar wind is, while the smaller the coronal hole, the fewer particles ejected, the milder the solar wind generated. Coronal holes refer to the dark regions of the Sun's surface. The sum of the pixel values of the dark part of the Sun surface is chosen as the feature representation of the coronal hole information. We first use the method of visual observation to determine the approximate range of coronal hole pixel values. Then, we conduct a number of comparative experiments, and find that when the threshold is set to 15, relatively accurate coronal hole information could be obtained, and the best results could be achieved. The experimental results are shown in Table 5.

4.2.4. Selection of Attention

In order to demonstrate the validity of our proposed TDAM, we conduct a number of ablation experiments. In rows 1–3 in Table 6, we list the results of no attention mechanism and single-dimensional attention mechanism. It can be seen that our proposed TDAM achieves the best results, which indicates both attention mechanisms are important for our model.

4.2.5. Selection of Decoder

After extracting the features from the two dimensions through the attention mechanism, we use RNN to predict the time series. We compare the simple RNN, LSTM, and GRU structures, respectively. As shown in Table 7, where 1RNN means that the decoder part is set to single-layer simple RNN, 2RNN means two-layer RNN, and the others are similar. Because the simple RNN cannot handle long-term dependence due to the phenomenon of gradient disappearance, the worst result is achieved. LSTM solves this problem by introducing gate mechanism, and the performance is improved. On the other hand, GRU achieves the best experimental result by combining the forget gate and input gate into a single update gate and is not easy to overfit, as shown in Table 7. The TDAM prediction model implements the lowest RMSE and MAE and the highest CC. The fitting of predicted value and observed value of the TDAM model is shown in Figure 8, and it can be seen that the real value is well fitted.

5. Discussion and Conclusion

In this study, TDAM is proposed to predict the solar wind speed 24 h in advance. It combines attention mechanism and GRU. The attention mechanism can allocate the importance of the time dimension and the feature dimension, respectively. Then GRU with excellent performance on time series prediction can capture long-term information.

In the stage of data analysis and processing, we first select seven input features: history speed, proton density, proton temperature, flow pressure, sigma-B, ICME, and coronal hole information. The first five features were taken from OMNI data. Since the solar wind speed is mainly affected by coronal holes and ICME, we introduce the ICME time list information to represent the features of ICME, and extract the features of coronal holes from the EUV filtergrams taken by SDO. We collect the data from 2011 to 2017 and divide them into training set, verification set and test set. Then the outliers of the seven selected features are processed

Table 7
Performance Comparison Between Different Encoder Structures and Two-Dimensional Attention Mechanism (TDAM)

Model_Structure	RMSE	MAE	CC
1RNN	68.430	53.612	0.750
2RNN	63.736	48.693	0.783
1LSTM	64.493	48.892	0.777
2LSTM	64.462	48.800	0.779
1GRU (our)	62.882	47.814	0.789
2GRU	65.727	49.960	0.768

Notes. Bold values are the best performance compared with the other methods in each table.

Abbreviations: CC, correlation coefficient; GRU, gated recurrent units; LSTM, long short-term memory; MAE, mean absolute error; RMSE, root-mean-square error; RNN, recurrent neural network.

by cubic B-spline curve interpolation, the data are normalized according to the mean and standard deviation. The period of the solar wind speed is calculated by Fourier transform, and the data are aligned by the zero-padding method. All the data analysis and processing in the early stage are helpful for us to construct the prediction model.

In Section 4, we compare the TDAM model with other prediction models according to the three performance indices of RMSE, MAE, and CC. In the comparison experiment, RMSE and MAE of the TDAM model are smaller than those of other models, and CC is larger than that of other models, achieving the best performance. Moreover, the relationship between the features and the solar wind speed has been proved by ablation experiments, and our model can learn this relationship well. In addition, the TDAM model proposed by us can not only capture the long-term dependence of the time dimension, but also pay attention to the features that have important influence on the results due to the addition of TDAM. The experiment results show that the proposed model is effective in improving the accuracy of solar wind speed prediction. By using the same data set, the comparison of our proposed model with other existing

models shows that this study is helpful. Moreover, our model can achieve good prediction without prior physical knowledge. Such a method can help us realize more tasks easily and quickly. The codes of TDAM and data used in this work are available from <https://github.com/syrGitHub/TDAM>.

In this work, we discuss the solar wind speed prediction model 24 h in advance and achieve good results, but the long-term prediction cannot be very accurate. A major issue for GRU is gradients exploding/vanishing. Besides, GRU easily uses a lot of memory to store the partial results for their multiple cell gates in the case of a long input sequence. Temporal convolutional network (TCN) (Bai et al., 2018) can solve these problems by setting different backpropagation path direction and making the filter shared across a layer. According to the analysis in Section 4.2.1, the short-term prediction mainly relies on the autoregressive characteristics of the solar wind speed itself. However, long-term prediction needs to be carried out in combination with the driving mechanism of the solar wind, and CME, activity area will also have an important impact. Therefore, using deep learning to extract features directly from EUV images can obtain more information, and the length of time to predict in advance also has effects. In the future work, we will consider using the original

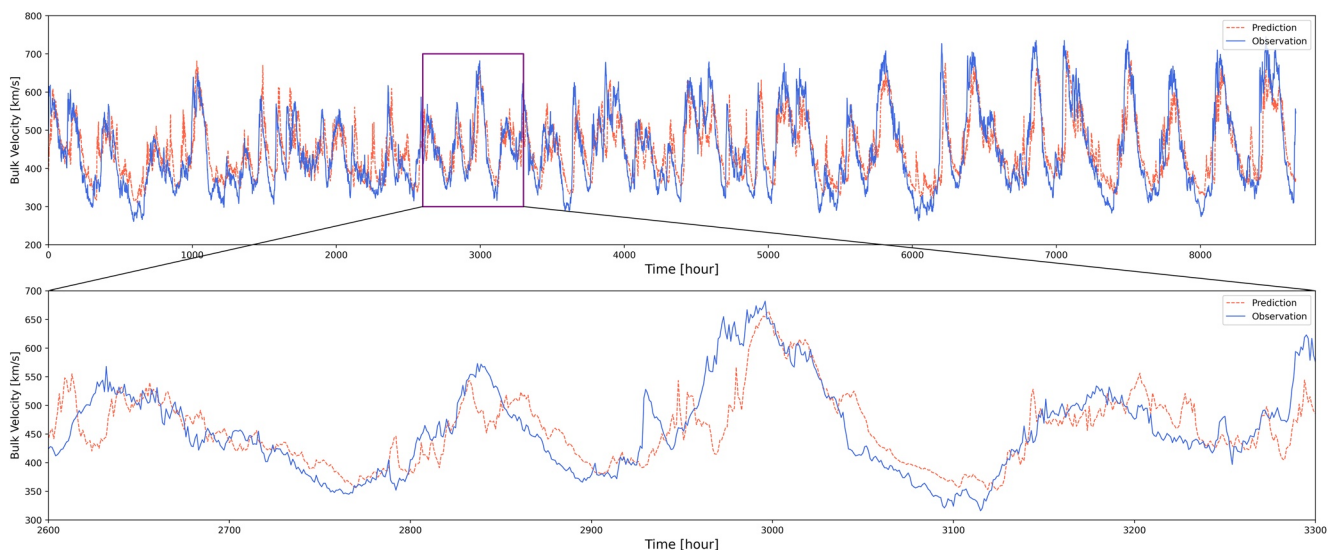


Figure 8. Fitting curve of the predicted value of the two-dimensional attention mechanism (TDAM) prediction model and the observed value of the verification set. The blue line is the observed value, and the red line is the predicted value of the TDAM model 24 h in advance. It is a good predictor of the upward and downward trends of observations, respectively.

images to directly capture the information of the solar surface for prediction and using the TCN to extend the prediction time, so as to further improve the accuracy of long-term prediction, and the data used in this study covers only one solar cycle. The performance of TDAM on other cycles needs to be studied in future.

Data Availability Statement

The codes of TDAM and data used in this work are available from <https://github.com/syrGitHub/TDAM>.

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